Qwen-Audio-3.0-Gen-Preview Technical Report

(Equal contribution; alphabetical by family name.)
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Abstract

Existing single-domain and multi-task audio systems remain limited in directly organizing speech, music, sound effects, ambience, and multiple roles into long-form temporal scenes. We present **Qwen-Audio-3.0-Gen-Preview**, a unified non-autoregressive framework that uses a Diffusion Transformer (DiT) and a shared variational autoencoder (VAE) to generate the complete mixed waveform. Prompt enhancement converts free-form requests into structured temporal records that are rendered as textual conditions, while a two-stage data curriculum and semantic conditional views train the proposed model to use these conditions across domains. A shared continuous VAE compresses 48 kHz stereo waveforms into 25 Hz latent sequences and incorporates semantic supervision, providing one representation for speech, music, sound effects, and their mixtures. On Seed-TTS-Eval, speaker similarity is the proposed model's clearest strength across all three subsets, and on the multi-speaker benchmark, the proposed model shows higher cross-turn consistency than Seed-Audio-1.0 in both languages. On AudioCaps, its advantages are concentrated in evaluations using large audio-language models and AudioBox. Relative to Seed-Audio-1.0, it achieves stronger temporal localization. Using approximately $0.1\times$ the music data of a dedicated in-house model, the proposed model remains close across all seven SongBench components and leads in three while retaining speech and general-audio capabilities. These results demonstrate the potential of unified generation for temporally structured, multi-domain audio.

1 Introduction

Existing audio generation models can already produce high-quality speech [4, 18, 47, 34, 3], general audio and sound effects [21, 22, 11], and music [6, 1] separately. However, real-world content creation often requires more than a single type of sound. A scene in a film, game, or podcast may contain character dialogue, background ambience, action-related sound effects, and music at the same time. Producing such a scene typically requires separate models for different audio components, followed by manual editing to place the generated clips on a timeline, align them, adjust their relative levels, and mix them into a final track. This multi-stage process is time-consuming and makes it difficult to maintain both overall coherence and precise control over individual audio components.

Recent work on unified audio generation has started to support speech, general audio, and music generation within a single framework [61, 45]. These models greatly expand the range of tasks that one model can handle. However, supporting multiple generation tasks is not the same as generating a unified audio scene. Existing models often generate speech, sound effects, or music in response to separate instructions, but pay less attention to how these elements should appear together, interact, and develop over time within one continuous track.

The key challenge of complex audio scene generation is not simply to expand the range of sounds a model can generate, but to organize heterogeneous sounds into a coherent scene. The model must control which sounds occur, when they start and end, how they overlap and follow one another, and how their relative levels are balanced. It must also coordinate characters, dialogue, ambience, sound effects, and music within a shared temporal structure. For long-form narratives, the model must further preserve character voices across dialogue turns and maintain continuity in the acoustic environment and background music over time.

In this report, we present **Qwen-Audio-3.0-Gen-Preview**, a unified non-autoregressive (NAR) framework that uses a Diffusion Transformer (DiT) operating in the continuous latent space of a shared variational autoencoder (VAE). Through a single generation path, the proposed model supports speech, music, sound effects, multi-speaker dialogue, and their temporally organized mixtures. Free-form user requests are converted into structured temporal conditions, while a two-stage training curriculum first establishes generation capabilities in individual audio domains and then teaches the proposed model to organize heterogeneous sounds into long-form scenes. To evaluate capabilities not well captured by existing benchmarks, we further introduce two in-house benchmarks for reference-conditioned multi-speaker speech generation and temporally controlled speech-and-sound generation. Across speech, general audio, and music, the proposed model achieves the highest speaker similarity among the evaluated systems on Seed-TTS-Eval, shows stronger cross-turn speaker consistency and temporal localization than Seed-Audio-1.0, and remains competitive with a dedicated music model while using only about $0.1\times$ as much music data. Together, these results show that a unified model can retain strong domain-specific generation capabilities while extending audio generation toward long-form, multi-character, and temporally structured scenes.

2 Related Work

2.1 From Domain-Specific Generation to Unified Audio Generation

Audio generation has traditionally been divided into separate tasks according to sound type. Text-to-speech (TTS) systems focus on speech intelligibility, naturalness, speaker similarity, and expressive control [4, 47, 18, 34, 3]. Recent systems have further extended TTS toward long-form and multi-speaker generation. VibeVoice supports long-form multi-speaker conversational synthesis through an ultra-low-frame-rate continuous speech tokenizer and next-token diffusion [30], while Any2Speech uses hierarchical annotations to model scene context, speaker profiles, utterance-level expression, and pronunciation [35]. Text-to-audio (T2A) systems generate environmental sounds and acoustic events from natural-language descriptions [21, 22, 11]. Text-to-music (T2M) systems further model musical elements such as melody, rhythm, instrumentation, vocals, and long-range structure, while supporting controls based on text or reference melodies [6, 1]. Although these domain-specific models achieve high generation quality on their respective tasks, they often rely on different audio representations, model architectures, and conditioning interfaces. When an application requires speech, sound effects, and music at the same time, several separate systems still need to be combined.

Unified audio generation aims to support multiple audio domains within shared representations or a shared model. UniAudio uses a unified discrete audio representation for a wide range of generation tasks [61], while Audiobox applies flow matching to both speech and general audio generation [45]. AudioX [41], UniFlow-Audio [59], UniSonate [31] and Audio-Omni [42] further expand the supported input modalities, generation tasks, understanding capabilities, and editing functions. UNISON also supports speech-and-sound generation, speaker cloning, temporal composition, and scene-level editing [20]. These models show that different audio domains can share representations and model parameters. However, their main focus is still to expand the range of tasks covered by a single model. A model may generate speech, sound, or music under different instructions, but may not organize them as connected parts of the same scene.

2.2 Complex Audio Scene Generation

Complex audio scene generation focuses on producing a complete mixed track that contains speech, sound effects, ambience, and music. Existing methods mainly follow two directions: external coordination and direct mixed-track generation with a single model.

External coordination methods first convert a user description into a script or timeline, then call several specialized models to generate different audio components, and finally combine them through post-processing. WavJourney uses structured scripts to organize speech, music, and sound generation [23]. Audio-Oscar further introduces multiple agents for sound design, timeline planning, model selection, generation, and post-production [9]. These methods are modular and allow direct editing of scripts and timelines. However, because different components are generated independently, overall consistency and cross-component interaction depend heavily on external mixing. Their multi-stage pipelines also increase system complexity and inference cost.

Another direction is to generate the final mixed track directly with a single model. Bagpiper can generate audio that combines speech, music, and sound effects using an autoregressive audio foundation model [40]. Foley-Omni focuses on generating complete soundtracks conditioned on video [37]. Dasheng AudioGen uses structured scene descriptions to represent speech, music, sound effects, and acoustic environments, and directly generates the mixture in a shared continuous semantic-acoustic latent space [26]. By modeling all components in one generation process, these methods can learn interactions between different sounds and avoid track-by-track generation and assembly. However, existing single-model approaches are still mainly designed for relatively short clips or simple combinations of sounds. Long-form scenes require stronger support for consistent character voices, multi-turn dialogue, and coordinated temporal structures across dialogue, ambience, sound effects, and music.

2.3 Temporal Control and Compositional Instruction Following

To improve controllability in multi-event audio generation, several studies have started to model the timing of sound events. Make-An-Audio 2 uses a large language model to convert natural-language descriptions into structured event sequences and supports variable-length generation [13]. PicoAudio and PicoAudio2 use temporally aligned supervision to control when events occur and how often they appear [56, 64]. ControlAudio combines text, timing, and phoneme conditions to improve both temporal accuracy and speech intelligibility [15].

Complementary efforts have focused on data construction, feedback-based training, and evaluation for temporal instruction following. AudioTime [55] and T2A-Feedback [48] evaluate event order, duration, frequency, temporal relations, and content completeness in multi-event audio generation. Together, these efforts broaden temporal instruction following from isolated event timing to compositional constraints over multiple events.

However, existing work on temporal control mainly focuses on general sound events and whether they occur at the specified time. Complex audio scenes involve a broader problem. The model must also handle character identity, multi-turn dialogue, persistent ambience, background music, and long-range relations among different components.

Rather than introducing another domain-specific generator or simply expanding the task coverage of a unified model, our work focuses on scene-level audio organization. Speech, sound effects, ambience, and music are modeled as interacting components of the same scene, together with their temporal relations, and jointly rendered into a complete mixed track in a shared continuous latent space.

3 System Overview

The proposed model provides a unified non-autoregressive generation path for speech, music, sound effects, and their mixtures. As illustrated in Figure 1, caption and text tokens jointly condition a Diffusion Transformer (DiT), which transforms an input sequence of noise latents into continuous VAE latents for the requested audio. For waveform reconstruction, the shared VAE decodes the continuous latent sequence generated by the DiT into the complete output waveform. All supported audio domains share the same conditioning interface, DiT, latent space, and VAE rather than relying on task-specific generation branches.

The same architecture is used throughout the two-stage training process. During pre-training, speech, music, and sound-effect samples establish basic acoustic generation quality and text–audio correspondence. During rich-timeline supervised fine-tuning, the proposed model learns from multi-character dialogue, persistent ambience, music, localized events, and heterogeneous mixtures, while replayed basic samples are used to mitigate loss of single-domain competence. The resulting system uses the same textual interface, DiT, and VAE for standalone speech, music, and sound-effect generation; multi-speaker conversations; speech with sound effects; and complete scenes containing dialogue, music, ambience, and temporally organized events.

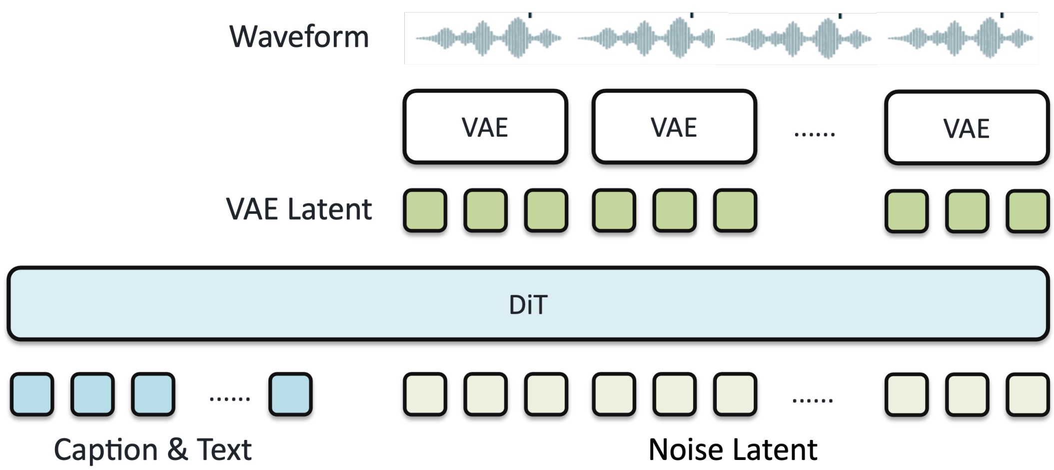

Figure 1: Overview of the unified non-autoregressive audio generation system. Caption and text tokens jointly condition a DiT that transforms a sequence of noise latents into continuous VAE latents. The shared VAE decodes the resulting continuous latent sequence into the complete output waveform. The same generation path models speech, music, sound effects, and their complex combinations.

4 Data

4.1 Data Curriculum

A two-stage data curriculum separates broad, single-domain acoustic learning from supervision for composing heterogeneous sources in time. Pre-training is used to establish competence in speech, music, and sound effects, while post-training—the rich-timeline supervised fine-tuning stage introduced in Section 3—teaches the proposed model how these sources coexist within one waveform. Figure 2 illustrates this design.

The pre-training stage covers three primary modalities: speech, music, and sound effects. Speech provides the dominant supervision mass and supports intelligibility, speaker attributes, paralinguistic variation, and background acoustics. Music contributes longer temporal structure, including rhythm, phrasing, instrumentation, and style continuity. Sound effects provide dense supervision for localized events, acoustic scenes, object interactions, and environmental cues. In addition to modality-level balance, we monitor duration buckets and semantic coverage during ingestion. Speech is tracked by language and speaker-related attributes; sound effects by event type and acoustic scene; and music by genre, mood, energy, instrumentation, and vocal presence.

The post-training stage focuses on organizing multiple sources in a shared acoustic scene. Its mixture emphasizes long-form dialogue, ambience-rich mixtures, music continuity, and explicit sound events. Long-form conversational speech provides supervision for speaker consistency and turn-taking; mixed-scene audio provides foreground/background supervision under realistic acoustic conditions; music is used to preserve long-range style and bed-generation ability; and sound effects provide supervision for localized timing and transient-event control. Table 1 summarizes the relative mixture design. To mitigate overfitting to noisy or post-produced media, we replay examples of clean speech, music, and sound effects during post-training. This replay is intended to preserve single-domain quality while retaining the post-training emphasis on compositional control.

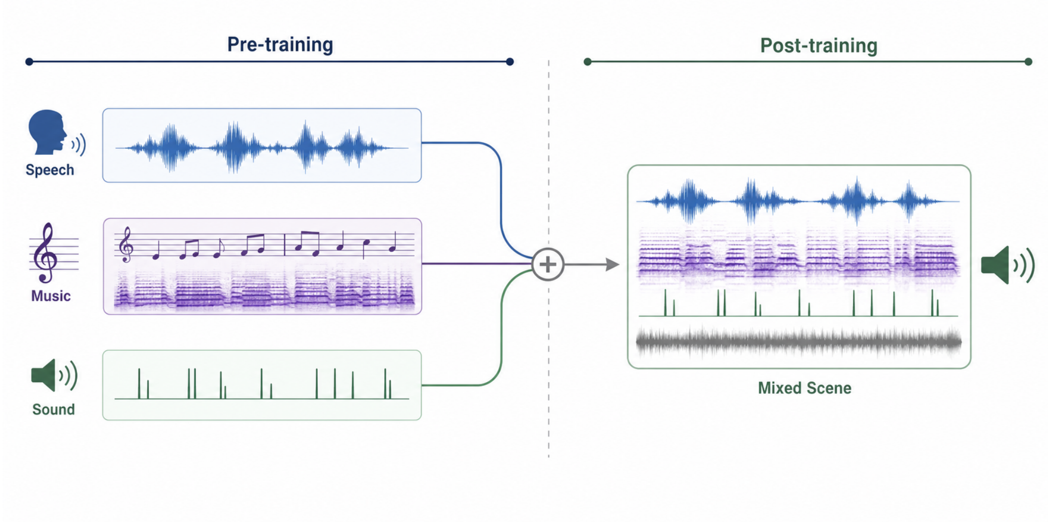

Figure 2: Overview of the two-stage data curriculum. Single-domain speech, music, and sound-effect supervision is first learned during pre-training, then combined during post-training to support coherent mixed-scene generation.

Table 1: Relative composition and role of the data curriculum. Pre-training shares are duration-normalized, and post-training shares denote mixture weights.

Data family	Relative share	Main role
Pre-training		
Speech	67.2%	Linguistic content, speaker attributes, paralinguistics, and background acoustics
Music	30.0%	Long-range musical form, rhythm, style, instrumentation, and vocal/instrumental attributes
Sound effects	2.8%	Event categories, acoustic scenes, event sources, and environmental context
Post-training		
Long-form speech	36.4%	Long-context dialogue, speaker consistency, and turn-taking
Mixed-scene audio	30.3%	Scene realism, ambience, and foreground/background organization
Music	30.3%	Music continuity, style control, transitions, and beds
Sound effects	3.0%	Foley, transient events, and localized acoustic actions

Overall, the curriculum progresses from learning what each sound is to learning how sounds coexist. This section defines the composition of the training data; the next section describes how heterogeneous examples are converted into structured annotations.

4.2 Data Annotation Pipeline

The data pools described above arrive with heterogeneous supervision. Clean speech may already provide transcripts, speaker identifiers, or language metadata; music data may include lyrics and catalog-level descriptors; and sound collections may contain event labels, captions, or acoustic-scene tags. Long-form media, in contrast, often contain several overlapping components but only partial source-side annotations. Our annotation pipeline consolidates the available supervision with audio-derived annotations instead of reducing every example to a single flat caption. It produces a structured record that separates persistent scene properties, source-level attributes, and temporally localized content; Section 4.4 later describes how this record is rendered as a textual condition for the proposed model.

Source normalization and temporal decomposition. We first normalize source-side annotations into a shared inventory of semantic fields while preserving the distinction between observed content and descriptive attributes. Transcripts and lyrics are treated as primary audible content, whereas language, vocal characteristics, acoustic environment, musical style, and production properties are represented as attributes. For long-form audio, annotation is performed on a shared time axis. Speech regions, music spans, persistent ambience, and candidate sound events are localized independently, so their intervals may overlap. This is important for mixed media, where a dialogue turn can coexist with background music, room tone, and a short foreground event.

Speech turns and role linking. Speech activity is segmented into utterances, and available transcripts or subtitles are aligned to the corresponding intervals. Speaker diarization then assigns utterances to local speaker tracks. Tracks attributed to the same source are linked across dialogue turns to form a stable role identifier. Vocal attributes are aggregated at the role level rather than predicted independently for each utterance. A role profile can include identity information, gender or voice class, age when reliably available, accent or dialect, and timbre descriptors. This aggregation reduces turn-level inconsistency and provides the anchors needed to associate every dialogue line with the correct voice over a long context.

Soundscape annotation. Persistent acoustic conditions are annotated separately from local events. At the clip or scene level, the annotation describes the environment, room tone, background activity, and spatial or production characteristics that remain meaningful across multiple intervals. For mixed media, music entry, exit, and transition regions may additionally be localized when they form part of the temporal organization of the scene. These entries describe the temporal placement of music within a mixed soundscape; the annotations below describe the internal organization of a music sample.

Music annotation. Music is represented as a hierarchy of temporally coordinated annotations rather than a flat set of tags. A single global description cannot simultaneously express periodic organization, harmonic evolution, sectional form, vocal performance, note events, and acoustic realization. The pipeline therefore covers three complementary views, using the names shown in the figure: the Full Mix for overall rhythmic, harmonic, and sectional analysis; the Vocal Stem for source-aware vocal information; and the Full Track for symbolic and learned representations. Their purpose is not to duplicate the waveform, but to expose structure at the temporal granularity at which it is meaningful. Figure 3 summarizes these views and their feature families.

The Full Mix view describes how a piece unfolds. Beat, downbeat, tempo, and meter provide a common rhythmic scaffold, supporting temporal alignment while distinguishing local event placement from longer-range pulse and metric organization. Chords and key capture local harmonic motion and global tonal context, which together characterize progression and harmonic consistency. Section annotations and boundaries describe long-range form, partitioning a song into coherent units and making transitions and recurring passages explicit. These annotations connect frame- and event-level observations to the larger musical context in which they occur.

The Vocal Stem view focuses on the singing voice. Source separation and quality screening are supporting operations rather than annotation targets themselves; they isolate reliable vocal regions. Continuous F0 records fine-grained pitch movement and expressive variation, whereas vocal MIDI supplies a compact note-event abstraction; together they connect performance detail with symbolic melody. Lyrics and language associate the vocal line with linguistic content, while an identity embedding describes voice consistency without requiring a named singer label. This combination supports alignment among melody, words, and vocal character across sections.

The Full Track view complements these interpretable features with symbolic and learned representations. Multi-instrument MIDI summarizes polyphonic content as structured note events, exposing relationships among musical parts that cannot be conveyed by global tags alone. Across the three views, beat, tempo, key, section, lyrics, and language are human-readable semantic annotations, whereas vocal and multi-instrument MIDI are symbolic representations. Discrete audio tokens and continuous VAE latents are learned acoustic representations that retain detail—including timbre, texture, and production characteristics—that cannot be exhaustively expressed by a hand-designed vocabulary. These learned representations are stored separately from the human-readable annotations and are not assumed to be serialized into natural-language templates. All streams share an audio time base and an explicit scope—track, section, event, frame, or source—so downstream training can

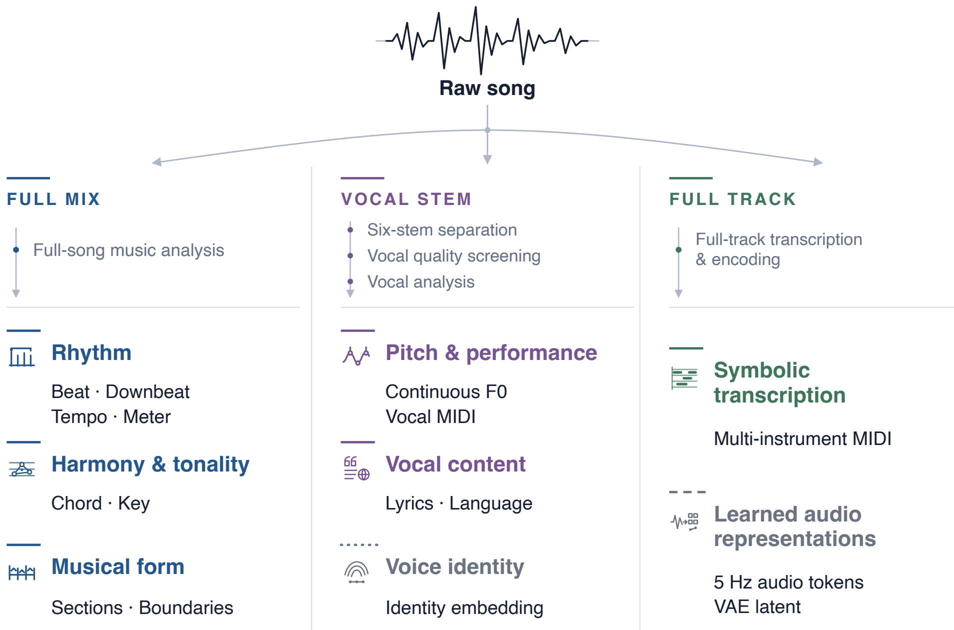

Figure 3: Multi-level music annotation pipeline. Full-mix features describe rhythmic, tonal, and structural organization; source-aware features describe vocal performance, linguistic content, and voice consistency; track-level representations provide symbolic event structure and complementary acoustic detail.

combine global organization, local performance, semantic content, and acoustic realization without collapsing their different meanings.

Localized non-speech events. Transient and independently meaningful sounds are retained as time-localized events even when they overlap with speech or music. Each event is associated with an interval or temporal anchor and, when supported by the audio, a description of the event, its acoustic source, and the underlying action. Event annotations are not absorbed into a dialogue caption or a global scene description. Repeated detections of the same acoustic occurrence are merged, while distinct overlapping events remain separate entries on the shared timeline.

Hierarchical assembly and consistency checks. The component annotations are finally reconciled into a structured record,

$$R = (G, \mathcal{P}, \mathcal{E}, U), \tag{1}$$

where G is the global scene and soundscape, $\mathcal{P}$ is an optional set of source or character profiles, $\mathcal{E} = \{e_i\}_{i=1}^{N}$ is a time-ordered sequence of local events, and U is the primary audible content. This assembly stage resolves duplicated or conflicting descriptions, binds dialogue turns to role profiles, and determines whether a sound is persistent context or a localized occurrence. We apply structural consistency checks before admitting an example to the rich-timeline pool. Every visible role identifier must resolve to one profile; dialogue, lyrics, and sound events must remain aligned to their audible intervals; event ordering must agree with the waveform; and overlapping components must not be removed merely because another source is dominant. Attributes that cannot be assigned to a reliable owner or temporal scope are omitted rather than attached speculatively.

Real recordings therefore produce the structured record R. The synthetic recipes in Section 4.3 are mapped to compatible records, and Section 4.4 organizes these records into the textual conditions used by the proposed model.

4.3 Synthetic Audio Data Construction

Although the annotation pipeline above recovers structured supervision from real recordings, such data rarely provide independent control over event count, onset, duration, overlap, relative level, and spatial attributes. We therefore construct a complementary synthetic corpus using a recipe-driven soundscape synthesis framework. The design builds on the controllable foreground/background composition paradigm introduced by Scaper [33], subsequently adopted by DESED to create strongly annotated synthetic soundscapes [44] and used by FUSS to construct source-preserving mixtures with simulated room responses [50]. Rather than rendering a scene as an indivisible waveform, we represent it as a collection of independently addressable sources. The recipe exposes explicit parameters for acoustic content, temporal organization, spatial rendering, and signal degradation.

Source inventory. The source pool comprises screened speech, singing and music, foreground sound effects, and ambient recordings. These signals are treated as authoring components: *foreground* and *background* describe a source's role within a particular recipe rather than an intrinsic property of the recording. Parameterized procedural synthesis supplies additional primitives, including sinusoids, frequency sweeps, clicks, metronomes, beeps, impulses, and band-limited noise. For rare or designed sounds, such as fictional machines or magical events, we layer physically interpretable recordings with procedural components and, when necessary, quality-screened candidates produced by audio generation models.

Scene specification and temporal scheduling. Each example is instantiated from a scene specification that assigns every source a semantic role, source identifier, onset, duration, and relative level. At the scene level, the specification defines event counts, ordering and overlap constraints, and target-to-background signal-to-noise ratios. Given a recipe and random seed, a deterministic scheduler realizes exact repetition patterns, silence intervals, tempo constraints, causal event chains, and long-form state transitions. The same representation supports object–material–action contrasts, dense polyphonic scenes, and multi-stage narratives without collapsing their constituent events into a single uncontrolled generation step.

Acoustic and channel rendering. Acoustic variation is applied independently to each source. Depending on the recipe, a source may be rendered with room or binaural impulse responses and conditioned on direction, distance, motion trajectory, occlusion, and propagation effects. Our parameterized spatial simulation follows the general design principles of SpatialScaper [32]. Device and channel conditions are implemented as ordered processing chains that may include band limitation, equalization, static or background noise, nonlinear distortion, resampling, codec processing, and packet-loss effects. Whenever available, we retain both the clean input and each processed source track in addition to the complete mixed waveform.

Controlled counterfactuals. We construct paired counterfactual examples by cloning a valid scene recipe, modifying exactly one factor, and freezing all remaining factors. An intervention may change the number of events, exchange their order, remove a target, insert an additional event, or alter material, direction, distance, or channel conditions. Each intervention updates the recipe and its condition before the corresponding target waveform is rendered, producing a new valid condition–target pair. The resulting pairs reduce nuisance variation and support controlled positive, negative, and hard-negative comparisons for individual control dimensions. This construction differs from the conditional dropout in Section 5.1, which hides condition content without changing the current sample's true event order or target audio.

Annotations, quality control, and limitations. For every rendered mixture, we retain the instantiated recipe, random seed, component tracks or processed source signals, and construction-time event annotations. The recipe and construction-time event annotations are mapped into the structured record defined in Section 4.2. We refer to their temporal boundaries as *construction-time strong labels*, since a boundary specified by a recipe need not coincide perfectly with the perceptually audible activity in the corresponding source recording. Quality control comprises checks of semantic content, speech transcripts when applicable, event presence and count, temporal order, spectrum, loudness, true peak, clipping, aliasing, discontinuities, and rendering artifacts, followed by listening review of sampled outputs. Loudness and true-peak measurements follow ITU-R BS.1770-5 [14]. Synthetic scenes complement rather than replace real recordings: residual context in source clips and mismatch between simulated and real acoustics remain important limitations, and any downstream benefit of synthetic data must be established empirically. Together, real annotations and synthetic recipes provide compatible structured records for training-side conditioning. Section 4.4 also converts inference-side user input into this shared format before rendering the textual condition for the proposed model.

4.4 Prompt Enhancement and Condition Rendering

Real-world user prompts are often free-form and may describe different audio generation scenarios, including character dialogue, standalone sound effects, instrumental music, and songs with vocals. Each scenario can also contain detailed requirements. For example, a dialogue request may specify the characters' identities, emotions, speaking rates, number of speakers, and background ambience. Directly using such prompts may cause the proposed model to miss important details or misunderstand the relationships among sound events. We therefore develop a prompt enhancement (PE) module that maps free-form user inputs to the structured fields required by the condition renderer while preserving the user's explicit requirements and the stated relationships among sound events.

We adopt a large language model to identify the primary task type of each user request and then organize the corresponding fields with speech, music, or sound-effect rules. Accompanying speech, music, ambience, and sound-event requirements remain part of the request; these rule sets organize the fields and do not restrict the final output to a single modality.

- **Speech generation:** The PE module organizes the scene, ambient sounds, character profiles, and dialogue content. A character profile may include the character's identity, approximate age, gender, narrative role, voice characteristics, emotion, and speaking rate. The module also makes temporal relationships explicit, such as whether dialogue and sound events occur in sequence, interrupt one another, or overlap. For every dialogue line or sound event, it estimates the start and end times using the surrounding context and, for dialogue, the speaker's emotion and speaking rate. This line-level timeline is intended to support timing control for multiple speakers and complex sound events.
- **Music generation:** The PE module first distinguishes instrumental music from songs with lyrics. For instrumental music, it organizes requirements such as genre, mood, tempo, instrumentation, and arrangement. For songs with vocals and lyrics, it additionally organizes vocal requirements, such as singer characteristics and performance style. It preserves lyrics explicitly when provided by the user. When the user provides only a lyric theme or partial lyrics, a large language model is used to supply the necessary song sections and performance information.
- **Sound-effect generation:** The PE module identifies the main sounds, their acoustic characteristics, the spatial setting, and how the sounds evolve over time. It also organizes the order and relationships of multiple sound events while avoiding characters or dialogue that the user did not request.

Condition rendering. The annotation pipeline, synthetic construction process, and PE module all produce records compatible with R in Equation 1. To form the input to the proposed model, we first select a surface template from a finite family and apply the renderer $\mathcal{T}$,

$$C_{\text{full}} = \mathcal{T}(R). \tag{2}$$

Given a structured record and the selected template, the renderer is deterministic. Its stable prompt grammar places the global scene and ambience before source or character profiles, followed by time-sorted dialogue and independent events. Transcripts remain explicit, while lyrics remain distinct from style and production attributes. C_{full} denotes the complete textual condition and may contain both a scene caption and primary text fields such as a transcript or lyrics.

The renderer also enforces a token budget, omitting low-priority detail by rule rather than truncating the tail of an arbitrary prompt. Variation within the finite template family changes surface wording without hallucinating conditions or changing the underlying record.

Validation and repair. The PE output is validated against the template rules required by the proposed model. The validation checks whether the user's requirements are preserved, the output format is complete, characters are correctly matched to their dialogue, sound events follow the intended order, and lyric and duration constraints are satisfied. If the rewritten result fails these checks, the module performs a targeted repair using the reported validation errors. After validation, the renderer returns the enhanced textual condition, including the estimated timeline, as input to the proposed model.

Training annotations, synthetic records, and PE outputs therefore use compatible condition formats. Section 5 describes how different conditional views are constructed from this format during training.

5 Conditional Training for Rich-Timeline Audio

5.1 Semantic Conditional Views

Rich-timeline training uses alternative condition views derived from the structured record in Equation 1 and the renderer in Section 4.4. This section specifies which semantic units remain visible in each view.

In rich-timeline audio, independently dropping individual fields can leave dialogue as the dominant explanation for the target audio and weaken the contrast supplied by scene conditions. We therefore apply dropout to structured semantic units before rendering:

$$C_M = \mathcal{T}(\mathcal{D}_M(R)), \qquad M \in \{\text{full}, \text{speech}, \text{scene}, \varnothing\}. \tag{3}$$

Here, $\mathcal{D}_M$ hides semantic units in R before $\mathcal{T}$ renders the condition. The full view retains every available condition. The speech view retains dialogue, speech-side attributes, and the identity information needed to interpret them. The scene view retains the persistent soundscape and independent sound events while removing dialogue and roles together. The empty view is the unconditional condition. The mixture over these views is configurable, but each selected view must remain an interpretable request for the target audio.

Fine-grained dropout follows the same rule. An independent event may lose its description or timestamp only if the remaining row still provides an interpretable event description or localization; otherwise the complete event row is removed. After any deletion, relations cannot refer to a removed event or source. Fields whose semantics cannot be separated safely remain in the full view. Conditional dropout hides only condition content: it does not rewrite transcripts or lyrics, permute the true event order, or change the target waveform.

5.2 Attribute Contrast and Role-Bundle Integrity

Attribute contrast and long-context role binding place different constraints on conditional dropout. We apply each rule only where the remaining condition is semantically interpretable.

Attribute contrast. Age, gender or voice class, accent, and dialect use a canonical attribute block with an elevated and configurable missing probability relative to decorative detail. Across repeated draws, an independently interpretable attribute can be present or absent for the same target audio. This contrast is intended to expose the attribute-specific conditional difference to CFG. Whenever an attribute is visible, it remains normalized, explicit, and unambiguous; the sampling distribution is configured to retain attribute-present examples as well as attribute-absent draws.

Role-bundle integrity. A rich-timeline role bundle that is referenced by visible dialogue cannot be partially deleted. Whenever a visible turn is labeled `roleX`, its role card retains every available core anchor: identity, gender or voice class, age when annotated, accent or dialect when annotated, and at least one timbre anchor. Optional personality or lengthy descriptive prose may be dropped. The role card and its dialogue turns remain atomically bound, so deleting a role bundle also removes its linked turns from the visible condition. When reference audio is present, the reference audio, its textual label, and the matching label in the target dialogue remain bound.

Thus, independently interpretable attributes can participate in attribute contrast, whereas a role bundle already referenced by dialogue can only be removed together with its linked turns.

Counterfactual construction in Section 4.3 creates a new valid condition–target pair; conditional dropout instead leaves the audible content of the fixed pair unchanged.

5.3 Classifier-Free Guidance

The semantic views in Section 5.1 provide the visible and empty conditions used for classifier-free guidance (CFG). At inference, CFG combines the corresponding conditional and unconditional vector fields as

$$v_{\mathrm{cfg}} = v_u + s\big(v_c - v_u\big), \tag{4}$$

where v_c is predicted from the visible condition, v_u is predicted from the empty condition, and s is the guidance scale. The difference $v_c - v_u$ represents the proposed model's learned response to information in the visible condition, and s weights that difference. In this formulation, guidance relies on contrasts learned from consistent views and does not itself supply role or soundscape information absent from the visible condition.

5.4 Shared Continuous Audio VAE

The conditional mechanisms described above operate on a continuous audio latent space, which is defined for all supported domains by the shared VAE described here.

Design motivation. Qwen-Audio-3.0-Gen-Preview covers speech, music, and general audio and therefore requires a shared latent representation that supports high-fidelity reconstruction across these audio domains. We compress 48 kHz stereo waveforms into a continuous latent sequence at 25 Hz, reducing the sequence length processed by the DiT. We first train the VAE for high-quality reconstruction and subsequently introduce auxiliary semantic supervision so that the latent representation contains semantic information and is intended to better support downstream audio generation.

Latent representation. Following the convolutional waveform autoencoder design of Stable Audio Open [10], our VAE maps each waveform to a 128-dimensional continuous latent sequence. The encoder parameterizes a diagonal Gaussian posterior, and the decoder reconstructs the waveform from a sampled latent:

$$\begin{aligned} q_\phi(\mathbf{z} \mid \mathbf{x}) &= \mathcal{N}\big(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag}\boldsymbol{\sigma}_\phi^2(\mathbf{x})\big)\,, \\ \widehat{\mathbf{x}} &= D_\theta(\mathbf{z}), \qquad \mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x}). \end{aligned} \tag{5}$$

The latent sample $\mathbf{z}$ is used for waveform reconstruction, whereas the posterior mean $\boldsymbol{\mu}_\phi(\mathbf{x})$ is used for semantic supervision.

Semantic supervision. The posterior mean is mapped by a lightweight projection module into the embedding space of a frozen Qwen2.5-3B base language model. The projected audio representations are combined with an instruction prompt, and the frozen language model supplies a next-token prediction objective over the target response. Gradients update the VAE encoder and projection module, while the language-model parameters remain fixed. The language model is used only as an auxiliary supervisor during VAE training and is absent from downstream audio generation.

Training objective. The VAE generator objective combines reconstruction, variational, adversarial, feature-matching, and semantic terms:

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \mathbb{I}_{\mathrm{adv}}\left(\lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}} + \lambda_{\mathrm{fm}}\mathcal{L}_{\mathrm{fm}}\right) + \mathbb{I}_{\mathrm{sem}}\lambda_{\mathrm{sem}}\mathcal{L}_{\mathrm{sem}}, \tag{6}$$

where $\mathcal{L}_{\mathrm{rec}}$ contains multi-resolution spectral reconstruction terms evaluated on stereo waveform representations, $\mathcal{L}_{\mathrm{KL}}$ regularizes the variational posterior, and $\mathcal{L}_{\mathrm{adv}}$ and $\mathcal{L}_{\mathrm{fm}}$ denote the adversarial and discriminator feature-matching objectives. The indicators $\mathbb{I}_{\mathrm{adv}}$ and $\mathbb{I}_{\mathrm{sem}}$ specify whether the corresponding objectives are active for the current training stage and sample.

VAE training stages. We adopt a progressive training strategy distinct from the proposed model's pre-training and post-training curriculum in Section 4.1. The VAE is trained on approximately 200,000 hours of in-house audio data, predominantly consisting of music. We first optimize the waveform VAE with reconstruction, KL, adversarial, and feature-matching objectives to obtain a high-fidelity acoustic representation. Starting from this checkpoint, semantic continuation adds the frozen language model and projection module. The discriminator is temporarily disabled during early continuation; reconstruction-only and semantically annotated samples are jointly used, with the semantic objective applied only to the latter. The discriminator is then reintroduced while the reconstruction and semantic objectives are jointly optimized. This schedule is designed to incorporate semantic information while limiting degradation of the acoustic representation learned during reconstruction training.

Section 6 first evaluates the complete system on speech, general audio, and music, and then isolates the VAE through reconstruction measurements and controlled downstream probes.

6 Evaluation

6.1 Text-to-Speech Generation

6.1.1 Zero-Shot Speech Synthesis on Seed-TTS-Eval

We first evaluate zero-shot speech synthesis on Seed-TTS-Eval [2], a widely adopted benchmark for assessing the intelligibility and speaker preservation of reference-conditioned TTS systems. For each test instance, the proposed model is provided with a reference utterance and a target transcript, and is required to synthesize the target content while preserving the voice characteristics of the reference speaker.

Table 2: Results on the Seed-TTS-Eval benchmark. WER/CER (%) measure content intelligibility, and SIM denotes WavLM speaker similarity on a $[0, 1]$ scale. Bold and underlined values indicate the best and second-best results among the evaluated models in each column, respectively. A dash indicates that a result is unavailable or not applicable. Beyond TTS marks support for audio generation, interaction, understanding, or editing beyond conventional speech synthesis.

Model	**EN**		**ZH**		**Hard-ZH**		**Beyond TTS**
	WER (%) ↓	**SIM** ↑	**CER (%)** ↓	**SIM** ↑	**CER (%)** ↓	**SIM** ↑	
Human	2.14	0.734	1.26	0.755	–	–	–
Autoregressive Models							
Seed-TTS [2]	2.25	0.762	1.12	0.796	7.59	0.776	–
Ming-UniAudio [60]	1.85	0.580	0.95	0.700	–	–	✓
MiMo-Audio-7B-Instruct [38]	5.37	–	1.96	–	14.14	–	✓
UniMoE-Audio [24]	1.90	0.560	<u>0.80</u>	0.650	–	–	✓
Step-Audio-2-Mini [51]	3.18	–	2.13	–	16.31	–	✓
Qwen3.5-Omni-Plus [39]	<u>1.26</u>	–	0.99	–	–	–	✓
VibeVoice-1.5B [30]	3.04	0.689	1.16	0.744	–	–	–
FireRedTTS-2 [54]	1.95	0.665	1.14	0.736	–	–	–
Qwen3-TTS-12Hz-1.7B-Base [12]	**1.24**	–	**0.77**	–	–	–	–
MiniMax-Speech [62]	1.90	0.738	0.99	0.799	–	–	–
VoxCPM2 [65]	1.84	0.753	0.97	0.795	8.13	0.753	–
CosyVoice3-1.5B [8]	2.21	0.720	1.12	0.781	**5.83**	0.758	–
Qwen-Audio-3.0-TTS [53]	1.54	0.762	0.84	0.792	7.00	0.768	–
Non-Autoregressive Models							
F5-TTS (32 NFE) [4]	1.83	0.647	1.56	0.741	8.67	0.713	–
F5R-TTS [36]	–	–	1.37	0.754	8.79	0.718	–
LongCat-AudioDiT-3.5B [57]	1.50	<u>0.786</u>	1.09	<u>0.818</u>	<u>6.04</u>	<u>0.797</u>	–
Qwen-Audio-3.0-Gen-Preview	1.61	**0.805**	1.06	**0.819**	8.25	**0.808**	✓

Qwen-Audio-3.0-Gen-Preview records the highest SIM on the EN, ZH, and Hard-ZH subsets, making speaker preservation its clearest numerical strength on this benchmark. Its WER/CER values are not the column minima, so the results do not establish a uniform advantage in linguistic accuracy. The proposed model also supports the broader Beyond TTS task range, showing that a unified generation scope is compatible with strong zero-shot speaker similarity without implying leadership on every TTS metric.

6.1.2 Reference-Conditioned Multi-Speaker Speech Generation

Seed-TTS-Eval primarily evaluates reference-conditioned generation for a single target speaker at a time. To assess the ability to jointly control multiple speakers, we construct an in-house benchmark for reference-conditioned multi-speaker speech generation.

The benchmark consists of 100 instances each in English and Chinese, drawn from the Seed-TTS test set [2], with 96 and 104 distinct reference speakers, respectively. Each instance contains K reference utterances from distinct speakers and a speaker-attributed target script $\{(s_i, t_i)\}_{i=1}^{N}$, where s_i indicates the intended reference speaker and t_i is the text of the i-th turn. The proposed model is required to render the entire dialogue as a single waveform while assigning each turn to the correct reference voice and preserving speaker identity across turns.

Table 3: Results on our reference-conditioned multi-speaker speech generation benchmark. WER and CER (%) measure content intelligibility over the complete dialogue waveform. SIM denotes WavLM speaker similarity between each generated segment and its reference audio. Consistency (CONS) denotes WavLM speaker similarity between segments of the same speaker across dialogue turns, reflecting voice preservation in multi-turn generation. Both similarity metrics use a $[0, 1]$ scale. Bold and underlined values indicate the best and second-best results in each column, respectively.

Model	EN			ZH		
	WER (%) ↓	**SIM ↑**	**CONS ↑**	**CER (%) ↓**	**SIM ↑**	**CONS ↑**
Seed-Audio-1.0	**1.20**	**0.575**	<u>0.659</u>	**1.35**	<u>0.661</u>	<u>0.704</u>
Qwen-Audio-3.0-Gen-Preview	<u>1.32</u>	<u>0.514</u>	**0.702**	<u>1.99</u>	**0.682**	**0.740**

Compared with Seed-Audio-1.0[1], Qwen-Audio-3.0-Gen-Preview records higher CONS in both languages and higher SIM in ZH. Seed-Audio-1.0 records lower EN WER and ZH CER, as well as higher EN SIM. Thus, the proposed model's consistent advantage across both languages lies in cross-turn speaker consistency, rather than a uniform lead in intelligibility or speaker fidelity.

6.2 Text-to-Audio Generation

6.2.1 Evaluation on AudioCaps

We evaluate text-to-audio generation on the AudioCaps test set [16], using the publicly accessible OpenSound release, which contains 4,411 test caption instances.[2] Because multiple audio realizations may satisfy a caption, we use complementary automatic alignment measures rather than direct waveform-reference metrics. All systems are evaluated using the same set of caption prompts. We report LAION-CLAP and MS-CLAP on the full test set to maintain comparability with established AudioCaps evaluations. We additionally employ Qwen3-Omni-30B-A3B-Instruct [58] as an audio-language judge over the full set.

To provide a complementary cross-model evaluation, we uniformly sample 100 prompts without replacement using a fixed random seed of 20260728. The same prompts and generated audio samples are evaluated using both CLAP metrics and three independent large audio-language model (LALM) judges: Qwen3-Omni-30B-A3B-Instruct, Qwen3.5-Omni-Plus[3], and Gemini-3.1-Pro-Preview[4]. The judges assess whether the generated audio satisfies the events, attributes, and temporal relations specified by the input caption. All baseline systems are reproduced following the settings described in their original papers, and the reproduced CLAP results are consistent with their reported performance.

In Table 4, Qwen-Audio-3.0-Gen-Preview leads the full-set Qwen3 column and the subset Qwen3 and Qwen3.5 columns, ranks second under Gemini, and does not lead any CLAP column. Its mean across the three subset judges is 0.8373, compared with the next-highest mean of 0.8177 from UniFlow-Audio; its full-set and subset Qwen3 scores differ by 0.0018.

[1] `https://docs.byteplus.com/en/docs/byteplusvoice/seedaudio-01`
[2] `https://huggingface.co/datasets/OpenSound/AudioCaps/viewer/default/test`
[3] `https://help.aliyun.com/en/model-studio/qwen-omni`
[4] `https://ai.google.dev/gemini-api/docs/models/gemini-3.1-pro-preview`

Table 4: Results on the AudioCaps benchmark. The full-set evaluation contains 4,411 caption prompts, while the fixed random subset contains 100 prompts uniformly sampled without replacement using seed 20260728. Qwen3 denotes Qwen3-Omni-30B-A3B-Instruct, Qwen3.5 denotes Qwen3.5-Omni-Plus, and Gemini denotes Gemini-3.1-Pro-Preview. All LALM columns report mean audio–text semantic-alignment scores. Bold and underlined values indicate the best and second-best results in each column, respectively.

Model	**Full Test Set** ($n = 4{,}411$)			**Fixed Random Subset** ($n = 100$)				
	CLAP ↑		**LALM ↑**	**CLAP ↑**		**LALM ↑**		
	LAION	**MS**	**Qwen3**	**LAION**	**MS**	**Qwen3**	**Qwen3.5**	**Gemini**
AudioX [41]	0.3208	0.4042	<u>0.7832</u>	0.3137	0.3980	<u>0.7995</u>	0.7000	0.8400
UniFlow-Audio [59]	**0.3712**	0.3924	0.7467	**0.3765**	0.3852	0.7245	<u>0.8100</u>	**0.9185**
Dasheng-AudioGen [26]	0.1969	**0.5086**	0.7327	0.2003	**0.5053**	0.7325	0.6790	0.7625
MMAudio [5]	<u>0.3612</u>	<u>0.4317</u>	0.7095	<u>0.3618</u>	<u>0.4416</u>	0.6775	0.7215	0.8375
Qwen-Audio-3.0-Gen-Preview	0.3178	0.3732	**0.8393**	0.3002	0.3549	**0.8375**	**0.8205**	<u>0.8540</u>

Table 5: Perceptual-quality results on the AudioCaps benchmark evaluated using the AudioBox metrics PQ, PC, CE, and CU. All systems are evaluated on the same test prompts. Bold and underlined values indicate the best and second-best results in each column, respectively.

Model	**PQ ↑**	**PC ↑**	**CE ↑**	**CU ↑**
AudioX [41]	5.711	<u>3.196</u>	<u>3.528</u>	4.969
UniFlow-Audio [59]	5.647	3.125	3.435	4.923
Dasheng-AudioGen [26]	<u>5.942</u>	2.948	3.493	<u>5.180</u>
MMAudio [5]	5.493	2.989	3.309	4.867
Qwen-Audio-3.0-Gen-Preview	**6.256**	**3.616**	**3.960**	**5.389**

The CLAP and LALM results exhibit different system-level rankings, reflecting their distinct evaluation mechanisms. At the individual-example level, the Spearman correlations between the three LALM judges and LAION-CLAP range from 0.136 to 0.254, while those with MS-CLAP range from 0.037 to 0.091. These low correlations support treating the LALM judgments as complementary to, rather than replacements for, the established embedding-based CLAP measures: the former directly assess fulfillment of events, attributes, and temporal relations in the caption.

Table 5 further evaluates perceptual quality using AudioBox [43]. Qwen-Audio-3.0-Gen-Preview records the highest values across all four dimensions. Taken together, the results locate its clearest numerical advantages in the LALM and AudioBox assessments rather than CLAP, and do not establish a uniform advantage across evaluation mechanisms.

6.2.2 Compositional and Temporal Instruction Following

Global embedding-based metrics on benchmarks such as AudioCaps provide an aggregate measure of audio–text alignment, but do not explicitly assess whether the requested events occur at the specified times. We therefore construct an in-house benchmark of 100 complex speech-and-sound scenes, with durations ranging from 12 to 35 seconds, to evaluate event coverage and temporal adherence. Each prompt specifies multiple speech and sound events together with their start and end times.

For each generated sample, Gemini-3.1-Pro-Preview and Qwen3.5-Omni-Plus independently determine whether each requested event is audible and estimate its temporal interval. We report the proportion of requested events judged audible (event recall), mean intersection over union (mIoU) between each reference event and its best-matched predicted interval, and the proportions of events reaching IoU thresholds of 0.3 and 0.5 in Table 6.

Table 6: Results on our in-house complex speech-and-sound benchmark for compositional and temporal instruction following under Gemini-3.1-Pro-Preview and Qwen3.5-Omni-Plus judges. All values are percentages. Bold and underlined values indicate the best and second-best results under each judge, respectively.

Model	Recall ↑	mIoU ↑	IoU@.3 ↑	IoU@.5 ↑
Gemini-3.1-Pro-Preview as Judge				
Seed-Audio-1.0	**98.77**	<u>38.48</u>	<u>56.17</u>	<u>38.27</u>
Qwen-Audio-3.0-Gen-Preview	<u>88.58</u>	**43.73**	**66.36**	**50.00**
Qwen3.5-Omni-Plus as Judge				
Seed-Audio-1.0	**97.84**	<u>37.36</u>	<u>56.79</u>	<u>36.73</u>
Qwen-Audio-3.0-Gen-Preview	<u>87.35</u>	**43.12**	**65.74**	**43.83**

Both judges produce the same ranking pattern. Qwen-Audio-3.0-Gen-Preview records mIoU values of 43.73 and 43.12 under Gemini-3.1-Pro-Preview and Qwen3.5-Omni-Plus, respectively, compared with 38.48 and 37.36 for Seed-Audio-1.0. The proposed model also records higher rates at both IoU thresholds, whereas Seed-Audio-1.0 has higher event recall; this is a descriptive coverage–localization trade-off, and no uncertainty estimates are reported.

6.3 Text-to-Music Generation

SongBench [52] assesses seven complementary components: melody, arrangement, and structure characterize composition and long-range organization; vocal, instrumental, and mixing assess source rendering and the integration of musical elements; and musicality provides a holistic perceptual assessment. The in-house music model [7] is a dedicated music pipeline whose music-token planner is an autoregressive (AR) hybrid language model, whereas Qwen-Audio-3.0-Gen-Preview is the unified non-autoregressive (NAR) DiT described in Section 3.

Table 7: SongBench [52] component scores for base pre-trained checkpoints evaluated on the same 20 examples. Relative music-data scales are 1.0× for the in-house model and 0.1× for the proposed model. Parentheses give scores reported in [7] for the final in-house system after reward-based post-training, including RL, on a separate 500-example benchmark; these italic values are not directly comparable and are excluded from the base-checkpoint ranking. Bold and underlined values indicate the best and second-best base-checkpoint scores in each component, respectively. Arr., Mus., Instr., and Struct. abbreviate arrangement, musicality, instrumental quality, and structure.

Model	SongBench ↑						
	Melody	Arr.	Mus.	Vocal	Instr.	Mixing	Struct.
In-house music model	**5.750** *(7.200)*	**6.467** *(7.388)*	<u>4.548</u> *(6.368)*	**5.208** *(7.623)*	<u>6.448</u> *(7.352)*	<u>5.820</u> *(7.305)*	**5.425** *(6.965)*
Qwen-Audio-3.0-Gen-Preview	<u>5.608</u>	<u>6.331</u>	**4.813**	<u>5.154</u>	**6.559**	**5.884**	<u>5.322</u>

The in-house model records higher values for melody, arrangement, vocal quality, and structure. Melody concerns continuity and development in the musical line, while arrangement and structure concern how instrumental parts and song sections unfold over time. The higher values on these three components are consistent with the sequential and hierarchical planning bias of the AR music-token planner: stepwise prediction may help maintain dependencies among successive musical tokens and longer song segments. The separate vocal-quality lead concerns singing and vocal rendering and is not attributed here to the same planning hypothesis.

The proposed model records higher values for musicality, instrumental quality, and mixing. Instrumental quality concerns non-vocal musical rendering, while mixing concerns the balance and integration of concurrently audible sources. The NAR DiT jointly refines a complete sequence of continuous VAE latents; the higher instrumental-quality and mixing values are consistent with, but do not establish, the hypothesis that joint latent refinement can support the integration of overlapping instruments, timbres, and levels. Musicality is a holistic and subjective component, so its higher score should not be attributed to a single architectural mechanism.

The comparison consequently yields no uniform ranking. With a checkpoint shared across speech, music, and sound effects and only $0.1\times$ the in-house music-data scale, the proposed model remains numerically close to the dedicated model in all seven components and leads in three. This result suggests competitive music capability for the proposed model at the evaluated data scale despite its broader scope. It is not a controlled AR-versus-NAR ablation: the systems also differ in training data, representations, objectives, and model scope. Moreover, the evaluation contains only 20 examples and reports no uncertainty estimates, so the differences should be interpreted descriptively.

6.4 VAE Component Evaluation

After the task-level evaluations above, we evaluate the VAE in Section 5.4 through waveform reconstruction and controlled downstream probes. The probes train task-specific generators against different target representations; they do not evaluate the complete Qwen-Audio-3.0-Gen-Preview system used in the preceding subsections.

Evaluation setup. For reconstruction, we compare DAC [17], EAR-VAE [46], SAME-L [29], Stable Audio Open [10], Semantic-VAE [27], LosaTok [63] and HoliTok [19]. Speech is evaluated on Seed-TTS EN and ZH [2], music on the Song Describer Dataset [25] and MuChin [49], and general audio on AudioCaps [16]. We group representations by latent frame rate into High-Rate (≥ 40 Hz) and Low-Rate (< 40 Hz) systems. In the reconstruction tables, SR and LR denote waveform sample rate and latent frame rate.

For downstream probes, we use an F5-style conditional flow-matching generator [4] on LibriSpeech-PC-200 [28] for TTS and a UniFlow-Audio-style generator [59] on a 200-sample text-to-music set. Within each task, every target representation uses the same generator architecture, training data, and optimization budget. We denote the reconstruction-only checkpoint as *Qwen-Audio-Gen-VAE (Acoustic)* and its semantically continued counterpart as *Qwen-Audio-Gen-VAE*; Sem. indicates whether the target representation uses semantic supervision.

Reconstruction quality. We use ViSQOL for perceptual quality; Mel and STFT distances for spectral fidelity; SI-SDR for signal-level reconstruction; PESQ and STOI for speech quality and intelligibility; and CCPC and ICPC for stereo phase coherence.

Qwen-Audio-Gen-VAE (Acoustic) achieves the best Low-Rate Mel and STFT distances on both music datasets and AudioCaps, while remaining competitive on speech despite its 25 Hz latent rate. HoliTok performs better on several speech metrics, which is consistent with our music-dominated training corpus, where speech accounts for only a small fraction of the data.

Semantic continuation improves ViSQOL on both speech sets, MuChin, and AudioCaps, while leaving the Song Describer result nearly unchanged. It introduces modest degradation in spectral and stereo-coherence metrics, but largely preserves the overall reconstruction quality.

Controlled downstream probes. TTS is evaluated with WER, speaker similarity (SIM), and UTMOS. Text-to-music is evaluated with OpenL3-FD, PaSST-KL, and four AudioBox scores: content enjoyment (AB-CE), production quality (AB-PQ), production complexity (AB-PC), and content usefulness (AB-CU).

Qwen-Audio-Gen-VAE (Acoustic) records better WER, SIM, and UTMOS than both the Mel spectrogram and Stable Audio Open targets. Relative to Qwen-Audio-Gen-VAE (Acoustic), semantic continuation lowers TTS WER and raises UTMOS but reduces SIM; Semantic-VAE and LoSATok have lower WER and higher SIM than Qwen-Audio-Gen-VAE. In the text-to-music probe, Qwen-Audio-Gen-VAE improves all six reported metrics relative to its acoustic checkpoint and records better values than LoSATok, although Stable Audio Open has the highest AB-PC score. Because the generator architecture, training data, and optimization budget are held fixed within each task, these differences concern target representations rather than the proposed model as a whole.

Table 8: Speech reconstruction on Seed-TTS EN and ZH. Bold and underlined values indicate the best and second-best Low-Rate results in each metric, respectively.

Model	Seed-TTS EN							
	SR	**LR**	**ViSQOL ↑**	**Mel ↓**	**STFT ↓**	**SI-SDR ↑**	**PESQ ↑**	**STOI ↑**
High-Rate								
DAC	44.1 kHz	86 Hz	4.3919	0.4693	1.0233	11.2306	3.7712	0.9691
EAR-VAE	44.1 kHz	43 Hz	4.2253	0.5532	1.0599	13.2584	3.4765	0.9660
SAME-L	44.1 kHz	43 Hz	3.9208	0.6918	1.2487	13.4237	2.7375	0.9535
Semantic-VAE	16 kHz	40 Hz	3.2171	0.7818	1.7031	9.6977	3.6097	0.9738
Low-Rate								
HoliTok	48 kHz	25 Hz	<u>4.3905</u>	**0.4248**	**1.0050**	9.3901	**3.9208**	**0.9783**
Stable Audio Open	44.1 kHz	21.5 Hz	4.2587	0.6587	1.1594	6.7421	2.5371	0.9264
Qwen-Audio-Gen-VAE (Acoustic)	48 kHz	25 Hz	4.2282	<u>0.5683</u>	<u>1.0669</u>	<u>12.0236</u>	3.6142	0.9687
Qwen-Audio-Gen-VAE	48 kHz	25 Hz	**4.3927**	0.5788	1.0796	**12.1295**	<u>3.6455</u>	<u>0.9708</u>

Model	Seed-TTS ZH							
	SR	**LR**	**ViSQOL ↑**	**Mel ↓**	**STFT ↓**	**SI-SDR ↑**	**PESQ ↑**	**STOI ↑**
High-Rate								
DAC	44.1 kHz	86 Hz	4.3189	0.4370	0.9466	12.3867	3.8113	0.9654
EAR-VAE	44.1 kHz	43 Hz	4.2215	0.4624	0.8864	14.8875	3.6955	0.9643
SAME-L	44.1 kHz	43 Hz	3.8982	0.5460	0.9724	14.9420	3.0809	0.9488
Semantic-VAE	16 kHz	40 Hz	2.9406	0.8306	1.8793	10.3511	3.6086	0.9695
Low-Rate								
HoliTok	48 kHz	25 Hz	<u>4.3436</u>	**0.3735**	**0.8591**	10.7590	**3.9951**	**0.9759**
Stable Audio Open	44.1 kHz	21.5 Hz	4.1953	0.6067	1.0325	8.4022	2.6192	0.9246
Qwen-Audio-Gen-VAE (Acoustic)	48 kHz	25 Hz	4.2460	<u>0.4917</u>	<u>0.9235</u>	**13.4252**	<u>3.8554</u>	0.9676
Qwen-Audio-Gen-VAE	48 kHz	25 Hz	**4.3535**	0.5105	0.9274	<u>13.4169</u>	3.8533	<u>0.9684</u>

Table 9: Music reconstruction on the Song Describer Dataset and MuChin. Phase-coherence values are unavailable for the mono-native LoSATok. Bold and underlined values indicate the best and second-best Low-Rate results in each metric, respectively.

Model	Song Describer Dataset							
	SR	**LR**	**ViSQOL ↑**	**Mel ↓**	**STFT ↓**	**SI-SDR ↑**	**CCPC ↑**	**ICPC ↑**
High-Rate								
DAC	44.1 kHz	86 Hz	4.1981	0.8718	1.3921	10.8695	0.7212	0.7256
EAR-VAE	44.1 kHz	43 Hz	4.0576	0.4589	0.9725	11.9368	0.7909	0.7231
SAME-L	44.1 kHz	43 Hz	4.0121	0.5038	1.1843	11.7872	0.7848	0.7151
Low-Rate								
LoSATok	16 kHz	25 Hz	2.2283	1.9373	3.4818	-31.7331	–	–
Stable Audio Open	44.1 kHz	21.5 Hz	4.0653	0.6009	1.2062	6.0569	0.7047	0.5708
Qwen-Audio-Gen-VAE (Acoustic)	48 kHz	25 Hz	**4.3010**	**0.4534**	**1.0752**	**11.4310**	**0.7900**	**0.6922**
Qwen-Audio-Gen-VAE	48 kHz	25 Hz	<u>4.2925</u>	<u>0.5184</u>	<u>1.1615</u>	<u>10.8145</u>	<u>0.7787</u>	<u>0.6812</u>

Model	MuChin							
	SR	**LR**	**ViSQOL ↑**	**Mel ↓**	**STFT ↓**	**SI-SDR ↑**	**CCPC ↑**	**ICPC ↑**
High-Rate								
DAC	44.1 kHz	86 Hz	4.1430	0.9624	1.6574	9.9117	0.6947	0.7033
EAR-VAE	44.1 kHz	43 Hz	4.0065	0.5829	1.2510	10.7052	0.7821	0.7024
SAME-L	44.1 kHz	43 Hz	4.0289	0.6971	1.7745	10.2689	0.7654	0.6880
Low-Rate								
LoSATok	16 kHz	25 Hz	2.2051	1.6715	2.8214	-32.1244	–	–
Stable Audio Open	44.1 kHz	21.5 Hz	4.0009	0.6942	1.4057	5.1473	0.6905	0.5428
Qwen-Audio-Gen-VAE (Acoustic)	48 kHz	25 Hz	<u>4.2828</u>	**0.4778**	**1.1315**	**10.2356**	**0.7785**	**0.6729**
Qwen-Audio-Gen-VAE	48 kHz	25 Hz	**4.3064**	<u>0.5335</u>	<u>1.2268</u>	<u>9.9425</u>	<u>0.7734</u>	<u>0.6641</u>

Table 10: General-audio reconstruction on AudioCaps. Bold and underlined values indicate the best and second-best Low-Rate results in each metric, respectively.

Model	**AudioCaps**					
	SR	**LR**	**ViSQOL ↑**	**Mel ↓**	**STFT ↓**	**SI-SDR ↑**
High-Rate						
DAC	44.1 kHz	86 Hz	4.2854	0.5346	1.1450	5.9002
EAR-VAE	44.1 kHz	43 Hz	4.1789	0.6078	1.1402	7.2334
SAME-L	44.1 kHz	43 Hz	3.9516	0.7012	1.3015	7.6653
Low-Rate						
LoSATok	16 kHz	25 Hz	3.1208	1.2495	2.5070	-34.7308
Stable Audio Open	44.1 kHz	21.5 Hz	4.1664	0.6731	<u>1.1796</u>	0.4401
Qwen-Audio-Gen-VAE (Acoustic)	48 kHz	25 Hz	<u>4.2538</u>	**0.6075**	**1.1708**	**6.5950**
Qwen-Audio-Gen-VAE	48 kHz	25 Hz	**4.3450**	<u>0.6411</u>	1.1849	<u>6.4642</u>

Table 11: Controlled TTS probe on LibriSpeech-PC-200 after 600k training steps. Bold and underlined values indicate the best and second-best results in each column, respectively.

Target representation	**Sem.**	**WER (%) ↓**	**SIM ↑**	**UTMOS ↑**
Mel spectrogram	N	19.69	0.466	1.769
Stable Audio Open	N	12.33	0.504	1.892
Qwen-Audio-Gen-VAE (Acoustic)	N	7.71	0.507	3.015
Semantic-VAE	Y	**2.96**	<u>0.594</u>	3.034
LoSATok	Y	<u>3.30</u>	**0.622**	**3.411**
Qwen-Audio-Gen-VAE	Y	4.08	0.488	<u>3.367</u>

Table 12: Controlled text-to-music probe after 150k training steps, evaluated on 200 samples. Bold and underlined values indicate the best and second-best results in each column, respectively.

Target representation	**Sem.**	**OpenL3-FD ↓**	**PaSST-KL ↓**	**AB-CE ↑**	**AB-PQ ↑**	**AB-PC ↑**	**AB-CU ↑**
Stable Audio Open	N	144.387	1.103	5.408	6.240	**5.381**	6.250
UniFlow-VAE	N	139.165	1.058	4.768	5.569	<u>5.322</u>	5.563
Qwen-Audio-Gen-VAE (Acoustic)	N	140.333	1.065	5.959	6.537	5.063	6.658
LoSATok	Y	<u>119.792</u>	<u>1.054</u>	<u>6.269</u>	<u>6.984</u>	4.901	<u>7.128</u>
Qwen-Audio-Gen-VAE	Y	**118.954**	**1.021**	**6.397**	**7.114**	5.260	**7.188**

7 Conclusion

Qwen-Audio-3.0-Gen-Preview unifies speech, music, sound effects, and their mixtures through a single non-autoregressive generation path. Its shared textual interface conditions a DiT in the continuous latent space of a common VAE, whose 48 kHz stereo representation operates at 25 Hz and is augmented by semantic continuation after reconstruction training. The VAE decodes the generated latent sequence directly into the complete mixed waveform. Real annotations, synthetic recipes, and prompt-enhanced requests use compatible structured records rendered as textual conditions, while semantic views, role-bundle integrity, and classifier-free guidance expose complementary information without breaking source–dialogue relations.

Across audio domains, the proposed model shows strong speaker similarity on Seed-TTS-Eval and higher cross-turn consistency than the compared system on the multi-speaker benchmark. On AudioCaps, its clearest advantages occur under LALM and AudioBox assessments while the temporal benchmark shows improved localization. Using $0.1\times$ the music-data scale of the dedicated in-house model, the proposed model remains close across all seven SongBench components and leads in three. Reconstruction measurements and controlled probes support the utility of the low-frame-rate VAE representation. Together, these results highlight the potential for temporally controllable generation across audio domains, with data-efficient music capability and no task-specific generation branches.

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